

Bounded Lyapunov-Based Controllers for Linear Systems

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1 Introduction

1.1 Lyapunov-Based Controllers

Lyapunov-based control is a control method that prioritizes closed-loop stability over anything else. Using Lyapunov stability theory at the core of its formulation, Lyapunov-based controllers seek to produce control inputs that guarantee that all the conditions for a stable system are met. On their own, Lyapunov-based controllers are unconstrained, so they won't inherently satisfy state and input constraints; however, due to the work of Sontag, bounded Lyapunov-based controllers exist, meaning that constraints can be more naturally brought into the controller formulation.

1.2 Bounded Controllers with Faults

Bounded Lyapunov-based controllers are built upon a modification of Sontag's formula for bounded controllers. For the nonlinear system with multiplicative actuator faults modeled as,

$$\dot{x} = \mathbf{f}(x) + \mathbf{G}(x)\Theta u, \quad (1)$$

the bounded Lyapunov-based controller, used by Xue and El-Farra in their work on optimal fault-tolerant control [1], is defined as,

$$\mathbf{u}(x) = -k(x, \Theta) [L_{\mathbf{G}}V(x)\Theta]^\top \doteq \mathbf{k}(x, \Theta) \quad (2)$$

$$k(x, \Theta) = \frac{\alpha(x) + \sqrt{(\alpha(x))^2 + (u_{max}\|\beta^\top(\Theta, x)\|)^4}}{\|\beta^\top(\Theta, x)\|^2 \left[1 + \sqrt{1 + (u_{max}\|\beta^\top(\Theta, x)\|)^2}\right]} \quad (3)$$

$$\alpha(x) \doteq L_{\mathbf{f}}V + \lambda V = \frac{\partial V}{\partial x} \mathbf{f} + \lambda V \quad (4)$$

$$\beta(\Theta, x) \doteq L_{\mathbf{G}}V(x)\Theta, \quad L_{\mathbf{G}}V = \left[\frac{\partial V}{\partial x} \mathbf{g}_1 \quad \cdots \quad \frac{\partial V}{\partial x} \mathbf{g}_m \right]. \quad (5)$$

The α term can be understood as the evolution of the Lyapunov function as the system evolves along its trajectory purely autonomously (there are no input dynamics in α). The β term can be understood as the additional change in the evolution of the Lyapunov function due to a given control input (there are no autonomous dynamics in β).

The notation $L_B A$ is the Lie derivative, a generalization of the directional derivative, which is,

$$L_B A = \frac{\partial A}{\partial x} \cdot B$$

You don't really need to specify that it's a dot product as it's widely understood to be a dot product, however it's an important detail to remember.

Sontag's formula allows it to be shown that, starting from a valid initial condition, there exists control inputs that satisfy input constraints while also meeting specific levels of stability. More explicitly,

$$\|\mathbf{k}(x, \Theta)\| \leq u_{max}, \quad \forall x \in \Psi(u_{max}, \Theta), \quad (6)$$

where Ψ is the set of all states whose stability behavior can be addressed with a corresponding control input,

$$\Psi \doteq \{x \in \mathbb{R}^n : L_{\mathbf{f}}V + \lambda V \leq u_{max} \|\Theta^\top L_{\mathbf{G}}V(x)^\top\|\} = \{x \in \mathbb{R}^n : \alpha(x) \leq u_{max} \|\beta^\top(\Theta, x)\|\}, \quad (7)$$

and that starting from an initial condition within the set Ω defined as,

$$\Omega \doteq \{x \in \Psi(u_{max}, \Theta) : V(x) \leq c\}, \quad (8)$$

where $c > 0$ is the largest number for which $\Omega(u_{max}, \Theta) \subset \Psi(u_{max}, \Theta)$, the evolution of the Lyapunov function is guaranteed to satisfy,

$$\dot{V}(x) \leq -\lambda V(x), \quad (9)$$

and $\lambda > 0$, acting as a controller tuning parameter—this means that the time evolution of the Lyapunov function is negative semidefinite (since $V(x)$ is positive everywhere except the origin, where $V(0) = 0$), implying the origin is asymptotically stable. This specific formulation of the bounded controller guarantees that, given the system dynamics, the control input bounds, and the magnitude of the fault, if the state lies within Ω , there will be a corresponding set of control inputs that will make the system stable.

1.3 A Point About Ψ and Ω

For my own sake, I'd like to make a point about the nuances behind the sets Ψ and Ω . Ψ and Ω are both crucial in determining the “stability region” in state space where, starting from a point within this stability region, your system is guaranteed a certain level of stability. However, Ψ is the set of all points where the evolution of the Lyapunov function is, in a sense, bounded by the control input. This does *not* mean that all states within Ψ are necessarily stable, as Lyapunov stability theory is sufficient but not necessary (satisfying Lyapunov stability requirements means the system is stable from the perspective of the theory, but not satisfying them does not mean the system is unstable). Also, if you are at the boundary of Ψ where $\alpha(x) = u_{max} \|\beta^\top(\Theta, x)\|$, then the system states are still bounded but locked into whatever set of non-equilibrated states this results in—they are stable in the sense of Lyapunov, but not asymptotically stable. This is where we need to define Ω .

Ω is the level set within Ψ that, at the cost of being more restrictive in terms of the allowable states, allows us to guarantee that all states within Ω are trapped within Ω , and that they are stable. This makes Ω an invariant set, meaning all states within Ω will stay in Ω as $t \rightarrow \infty$. Ω being an invariant set allows us to work entirely within a region of stability guarantees, which then makes the math work out better.

That being said, Ω is not the end-all, be-all set where being outside of Ω means you are doomed to instability. It is entirely possible, and dare I say almost guaranteed, that there exist regions outside of Ω that also satisfy the same stability requirements (and this is especially true of nonlinear systems). However, the construction of Ω above is a conservative estimate on the stability region, and is just one of potentially several ways to define this stability region. Pick your favorite way of defining this stability region and you may find an even more accurate definition of Ω . Keep in mind that this specific foray into optimal bounded controllers isn't focused on the most accurate formulation of Ω , rather the broader picture once a suitable Ω is found.

2 Linearizing Stability Criteria

Our goal for right now is to see what the Ψ and Ω regions look like for a continuous LTI system. To do this, we need to restrict our scope from general, nonlinear systems to linear systems. To do this, we need to represent the Lie derivatives, Lyapunov function, and subsequently the α and β terms to be for linear systems. To ground ourselves, the autonomous LTI system we are working with is,

$$\dot{x}(t) = Ax(t) \quad (10)$$

2.1 Lyapunov Function and its Time Derivative

The Lyapunov function we will use isn't necessarily restricted to linear systems, but it's easy to use and satisfies all the requirements of a Lyapunov function:

$$V(x) = x^\top Px \quad (11)$$

where P is the solution to the continuous Lyapunov equation,

$$AP + PA^\top + Q = 0 \quad (12)$$

and $Q \succ 0$, where typically a good choice is $Q = I$. P is a symmetric matrix, which can make simplifications easier down the line. The time derivative of the Lyapunov function is commonly known,

$$\dot{V}(x) = \frac{\partial V}{\partial x} \cdot \frac{\partial x}{\partial t} = \left[\frac{\partial}{\partial x} (x^\top Px) \right] \cdot \dot{x}(t)$$

where we know that $\partial_x(x^\top Px) = 2Px$, and $\dot{x}(t) = Ax(t)$, so the time derivative is,

$$\dot{V}(x) = (2Px) \cdot (Ax) = x^\top (2PA)x. \quad (13)$$

2.2 Lie Derivatives

We now can linearize the Lie derivatives. The Lie derivatives we care about are,

$$L_{Ax}V = \frac{\partial V}{\partial x} \cdot Ax \quad \text{and} \quad L_BV = \frac{\partial V}{\partial x} \cdot B$$

which evaluate to,

$$L_{Ax}V = (2Px) \cdot Ax = x^\top (2PA)x, \quad (14)$$

and

$$L_BV = (2Px) \cdot B = x^\top (2PB) \quad (15)$$

2.3 α and β Terms

With the Lyapunov function and Lie derivatives linearized, we can now linearize the α and β terms. α is defined as,

$$\alpha(x) = L_{Ax}V + \lambda V,$$

which, linearized, is,

$$\boxed{\alpha(x) = x^\top (2PA)x + x^\top (\lambda P)x = x^\top (2PA + \lambda P)x}. \quad (16)$$

β is defined as,

$$\beta(\Theta, x) = L_BV\Theta,$$

which, linearized, is,

$$\boxed{\beta(\Theta, x) = x^\top (2PB)\Theta}. \quad (17)$$

2.4 Stability Regions Ψ and Ω

We can, finally, represent Ψ and Ω in terms of our linearized model. Ψ is defined as,

$$\Psi = \{x \in \mathbb{R}^n : \alpha(x) \leq u_{max} \|\beta^\top(\Theta, x)\|\},$$

which, linearized, is,

$$\boxed{\Psi = \{x \in \mathbb{R}^n : x^\top (2PA + \lambda P)x \leq u_{max} \|\Theta^\top (2PB)^\top x\|\}}. \quad (18)$$

Ω is defined as,

$$\Omega = \{x \in \Psi(u_{max}, \Theta) : V(x) \leq c\},$$

which, linearized, is,

$$\boxed{\Omega = \{x \in \Psi(u_{max}, \Theta) : x^\top Px \leq c\}} \quad (19)$$

and $c > 0$ is the largest number that makes $\Omega(u_{max}, \Theta) \subset \Psi(u_{max}, \Theta)$. I believe it might also be equivalent to say that $c > 0$ is the largest number that makes $\Omega(u_{max}, \Theta) \cap \Psi(u_{max}, \Theta) = \Omega(u_{max}, \Theta)$, which is one way of defining a positively invariant set.

To find Ψ all we'd have to do is find all the points in state space that satisfy its inequality. To find Ω , we just need to sample all the points within Ψ that satisfy its inequality (given that we can find a $c > 0$ that makes Ω positively invariant).

3 Model Used in Literature: Unstable Non-Isothermal CSTR

3.1 Verifying Unstable Equilibrium Point

For this attempt at bounded Lyapunov-based controllers, I will use the model used in Xue's paper: a non-isothermal CSTR containing the reaction $A \rightarrow B$, where the inlet concentration is the manipulated variable and the equilibrium point is the unstable equilibrium found in exothermic reactions:

$$\frac{dC_A}{dt} = \frac{F}{V}(C_{A0} - C_A) - k_0 e^{-E/RT} C_A \quad (20)$$

$$\frac{dT}{dt} = \frac{F}{V}(T_{A0} - T) - \frac{\Delta H}{\rho C_p} k_0 e^{-E/RT} C_A + \frac{Q}{\rho C_p V}. \quad (21)$$

The process parameters for this system are tabulated below:

Parameter	Value	Units
F	3.34×10^{-3}	m^3/h
V	0.1	m^3
k_0	1.2×10^9	h^{-1}
E	8.314×10^4	kJ/kmol
R	8.314	$\text{kJ}/\text{kmol}/\text{K}$
ρ	1000	kg/m^3
C_p	0.239	$\text{kJ}/\text{kg}/\text{K}$
ΔH	-4.78×10^4	kJ/kmol
C_{A0}^s	0.79	kmol/m^3
T_0^s	352.6	K
Q^s	0	kJ/h

with the equilibrium point $C_A^s = 0.577 \text{ kmol}/\text{m}^3$ and $T^s = 395.3 \text{ K}$. The first thing I want to do is make sure that the equilibrium point at $[C_A^s, T^s]$ is indeed at the unstable equilibrium point. Breaking out our reaction kinetics notes, we recall that the equilibrium points can be found in terms of the conversion X , where the conversion from the mole balance alone is governed by one expression, and the conversion from the energy balance alone is governed by another expression. Finding where the two conversions are equal yields the equilibrium points. We consider the mole balance and energy balance conversion expressions,

$$X_{MB} = -\frac{r_A V}{F} \quad (22)$$

$$X_{EB} = \frac{\sum_i^N (\theta_i C_{pi})(1 + \kappa)(T - T_c)}{-\Delta H} \quad (23)$$

where,

$$\kappa = \frac{U A}{F \sum_i^N (\theta_i C_{pi})} \quad (24)$$

$$T_c = \frac{\kappa T_a + T_0}{1 + \kappa}, \quad (25)$$

however, since the paper does not give us a U , A , or T_a , we can make due by overlaying the given equilibrium point on top of the mole balance and see where it lands. Using our knowledge that the reaction is first-order

in A , we can find that the mole balance expression is,

$$X_{MB}(T) = \frac{k_0 C_{A0} e^{-E/RT}}{\frac{F}{V} + k_0 C_{A0} e^{-E/RT}}, \quad (26)$$

and plotting this over a range of temperatures yields figure (1) below.

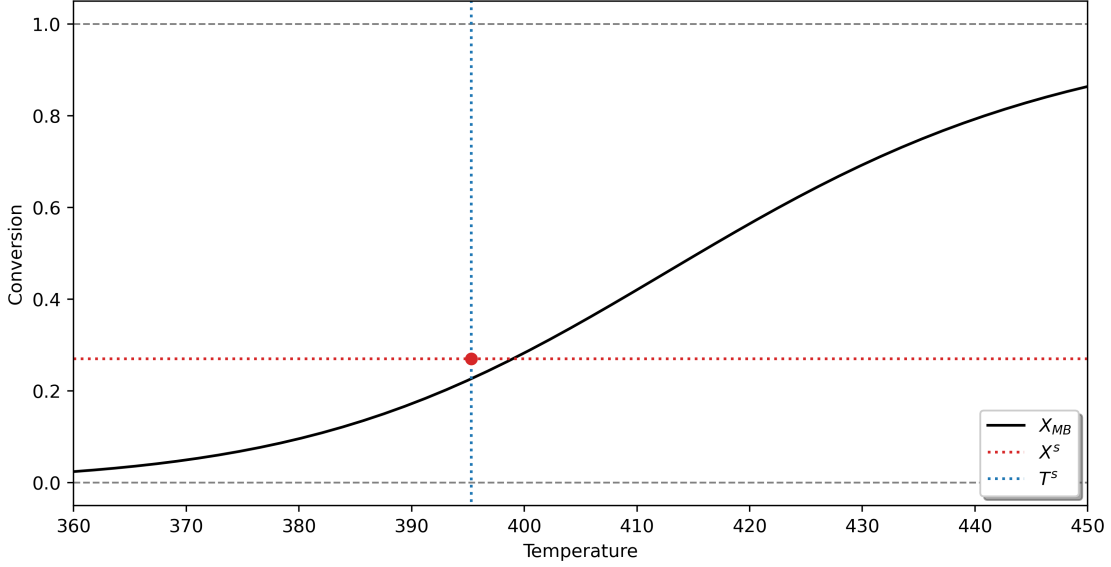


Figure 1: The given equilibrium point is close to intersecting the conversion profile, though not exact. It still provides useful information for modeling the system.

We can see that the given equilibrium point (the red dot) does not lie on top of the mole balance, so it is not an exact equilibrium point for the CSTR system, however we can still verify the equilibrium point by simulating the dynamic model and see how it evolves when the control input (again, the inlet concentration) is kept at the equilibrium input value given in the paper. These profiles are seen in figures (2) and (3).

This simulation verifies that the unstable equilibrium point chosen in the paper is actually quite close to the real unstable equilibrium point, so we can continue using the values in the paper in good faith.

3.2 Linearizing the Model

As usual, to get around dealing with nonlinear models, we will linearize this one around the equilibrium point. I want to explicitly write it out here because I want to minimize any errors that arise from using numerical linearization in Python. The LTI state-space model, again, is,

$$\dot{x} = Ax + Bu \quad (27)$$

where,

$$A = \begin{bmatrix} -\left(\frac{F}{V} + k_0 e^{-E/RT^s}\right) & -\frac{k_0 E}{R(T^s)^2} e^{-E/RT^s} C_A^s \\ -\frac{\Delta H}{\rho C_p} e^{-E/RT^s} & -\left(\frac{F}{V} + \frac{E \Delta H}{R(T^s)^2 \rho C_p} k_0 e^{-E/RT^s} C_A^s\right) \end{bmatrix} \quad (28)$$

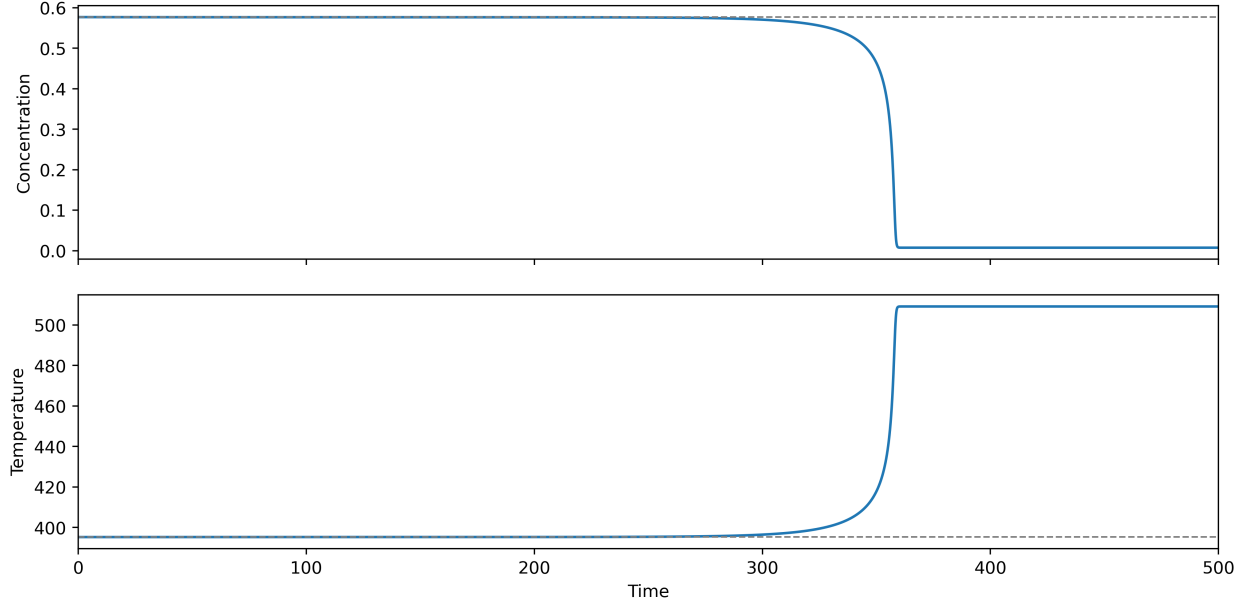


Figure 2: Dynamic response of system starting at initial condition $C_{A0} = 0.577 \text{ kmol/m}^3$ and $T_0 = 395.262534 \text{ K}$. The system stays at an unstable equilibrium point for quite a long time before veering off to a stable equilibrium point that yields a lower concentration of A .

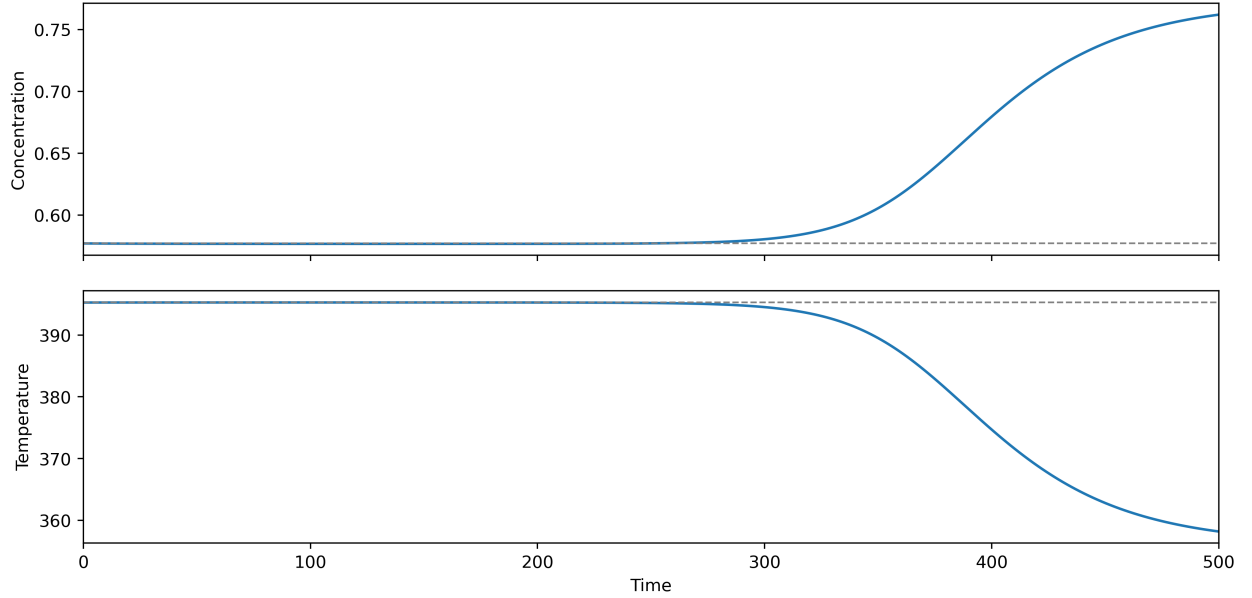


Figure 3: Dynamic response of system starting at initial condition $C_{A0} = 0.577 \text{ kmol/m}^3$ and $T_0 = 395.262533 \text{ K}$ just 0.000001 K difference from the other simulation. The system again stays at an unstable equilibrium point for quite a long time before veering off to a stable equilibrium point that yields a higher concentration of A .

and,

$$B = \begin{bmatrix} \frac{F}{V} \\ 0 \end{bmatrix} \quad (29)$$

Plugging in the process parameters, we end up with,

$$A = \begin{bmatrix} -4.578 \times 10^{-2} & -4.571 \times 10^{-4} \\ 2.063 \times 10^{-9} & 5.803 \times 10^{-2} \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 0.0334 \\ 0 \end{bmatrix}.$$

To verify that this autonomous system is at an unstable equilibrium, we can check that at least one eigenvalue in A is positive—the eigenvalues are $\lambda_1 = -0.0458$ and $\lambda_2 = 0.058$, meaning that the equilibrium point is indeed unstable.

3.3 Finding Ψ and Ω

To keep our results as similar to the paper as possible, we will use the same P , λ , and the input bounds u_{max} . Figures (4), (5), and (6) show some results. The dark blue regions are where the inequalities for both Ψ and Ω holds.

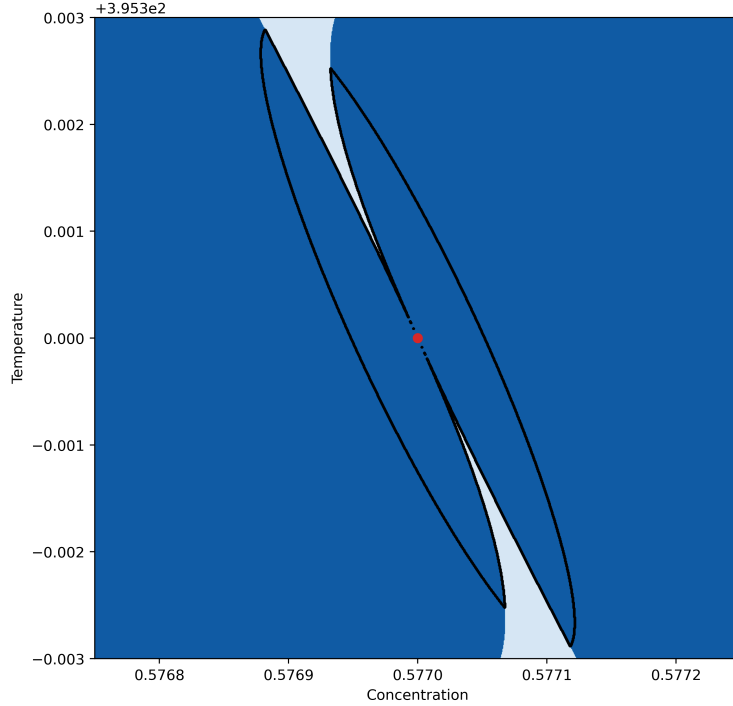


Figure 4: $\Theta = 1$

The figures show that there does not appear to be any radius around the equilibrium point such that Ω is not an empty set—there are two lobes that touch right where the equilibrium point is. The black ellipse was supposed to be Ω , but I just ended up using it to help me visualize the meeting point between the lobes. Changing the values of λ , Θ , and u_{max} do not change the fundamental shape of these two lobes, it just makes them larger or smaller. The only thing that really changes the shape of the lobes is the value of P , which you may know does not have a single “best” value. Depending on your choice of Q , assuming that it meets the requirement of being a symmetric matrix (or more correctly, a Hermitian matrix), P can take on the value that solves the Lyapunov equation, no other constraints. The shapes of these surfaces is quite interesting to me, and why they don’t even really resemble those in Xue’s paper is puzzling. In the next

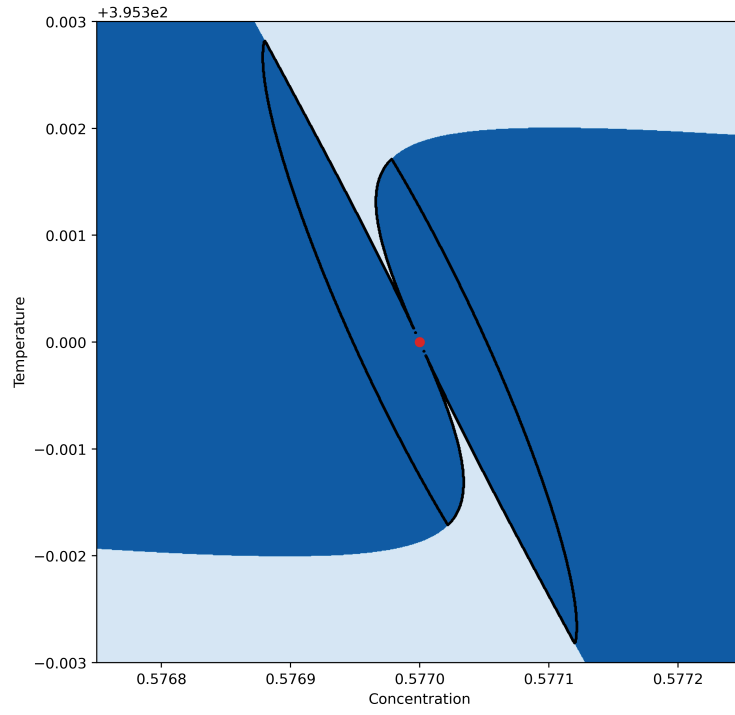


Figure 5: $\Theta = 0.5$

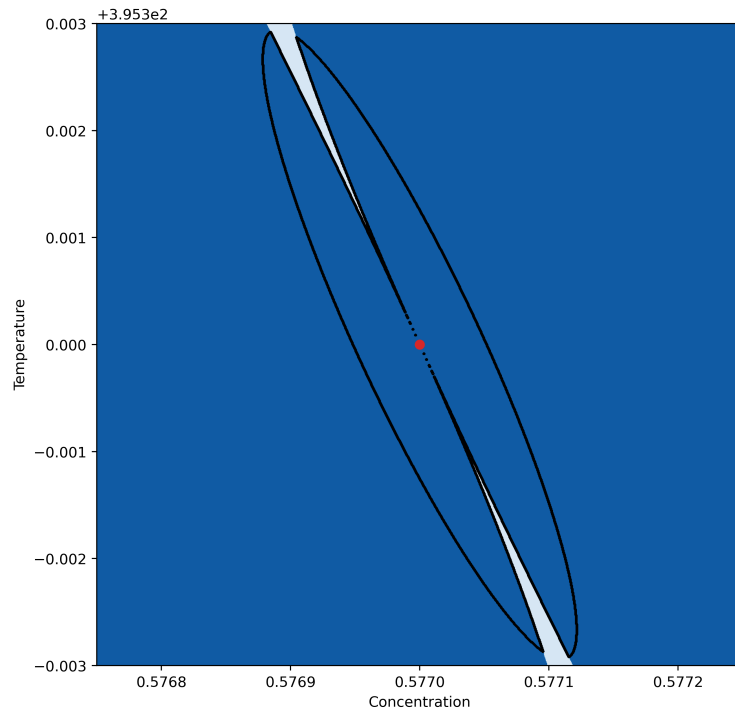


Figure 6: $\Theta = 2$

section I'll hope to explore even a tiny bit of why this might be so.

4 Exploring the Shapes of Ψ and Ω

The set that forms the most puzzling shape is of course Ψ , where instead of forming a well-defined region around the equilibrium point like in Xue’s paper, there is instead a single point at the equilibrium where two lobes meet, meaning that there cannot exist the set Ω .

In my rationalization of this, I attempted to first understand the inequality used to define Ψ :

$$x^\top(2PA + \lambda P)x \leq u_{max}\|\Theta^\top(2PB)^\top x\|$$

The left side of this inequality is a quadratic function, and the right side is a linear function—it must be noted that both sides of this inequality are zero at the origin and nonzero everywhere else.

Based on the properties of quadratic and linear functions, there will *always* be a portion of a quadratic function that is less than a linear function. Given any positive, nonzero slope of a line that crosses the origin, any quadratic function will have a region where it’s value is less than the straight line.

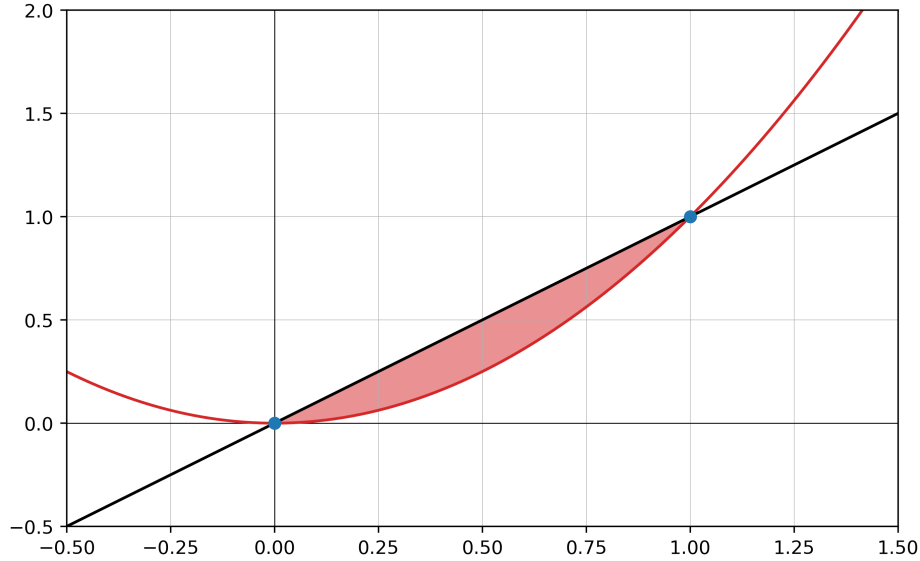


Figure 7: There is always a region where a (upwards-facing) quadratic function will be less than a linear function.

However, notice that the right side is not an ordinary linear function, and it’s actually the absolute value of a linear function. This means that there are two distinct regions where the quadratic function is less than the linear function.

We must notice that since both functions are zero at the origin, this forms a single point where the two lobes are disconnected. There is no way for the two lobes to become connected if both functions are equal at the origin. This context is important because the $\alpha(x)$ and $u_{max}\|\beta^\top(\Theta, x)\|$ surfaces that are formed within the state space, of course, are quadratic and linear functions, respectively, and are shown in figures (9a) and (9b) below.

From this, we can see why the inequality only forms two essentially disconnected lobes for the given system, where it is impossible to form an Ω level set around the origin, as that is exactly where the two sides of the inequality are equal. For there to be a region for Ω to exist, we would need to modify something

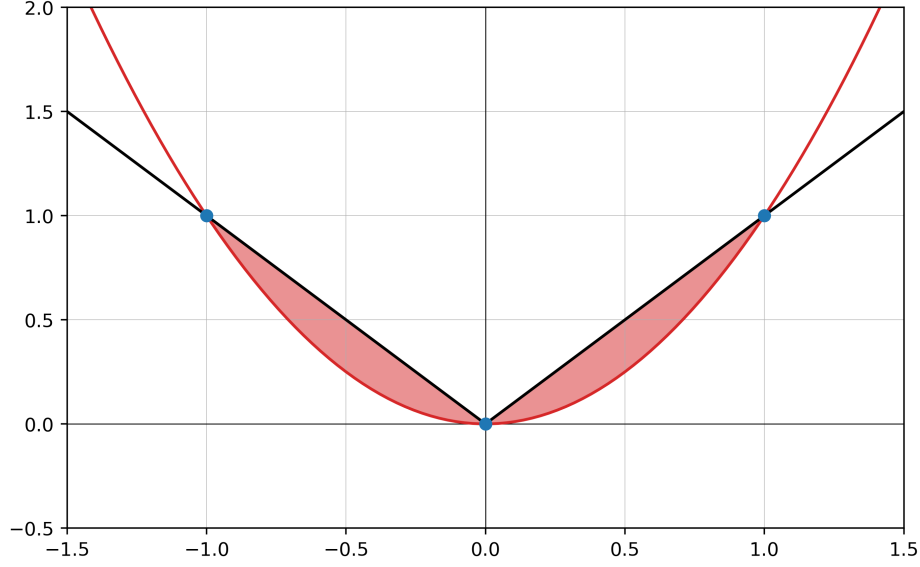
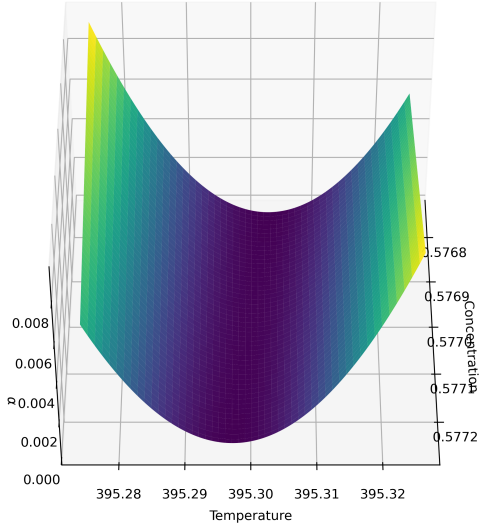
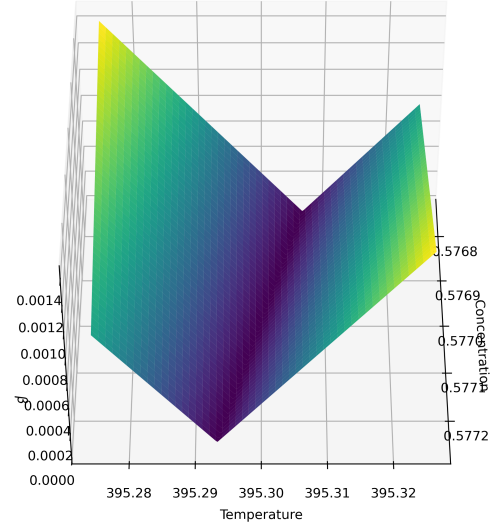


Figure 8: When you take the absolute value of a linear function, it now produces two lobes where the quadratic function is less than the linear function.



(a) The $\alpha(x)$ surface



(b) The $u_{max} \|\beta^\top(\Theta, x)\|$ surface

Figure 9: The $\alpha(x)$ and $u_{max} \|\beta^\top(\Theta, x)\|$ surfaces form a quadratic and absolute-valued linear surface, respectively.

about the system, such as the solution to the Lyapunov equation P (which can be directly chosen by us, or we can solve for P with different Q s). There must be some connection to having the $\alpha(x)$ inequality be a positive definite/negative definite/semi-definite matrix and the region for Ω to form, however I'm skeptical of this because no matter what you choose for P or Q or λ , the origin will always evaluate to zero. There must be some additive term we can add to either side that makes the math work out correctly, yet also gives us room to form Ω .

5 Conclusions

So not much was really found in this document, but it at least highlighted an issue we need to find a way around: the inequality that defines Ψ is one such that the origin is where two “lobes” meet, and Ω is an empty set. Likely due to the linearization, this causes a problem for us because Ω is the region where we eventually end up wanting to operate within. To get around this, we may need to look into reformulations of Sontag’s formula or different definitions of Ψ , perhaps adding an additive term to expand the region around the origin, or changing something to shift the meeting point for the lobes so that the origin is included within one lobe, and Ω can be formed within it.

References

- [1] D. Xue and N. El-Farra, “Forecast-triggered model predictive control of constrained nonlinear processes with control actuator faults,” *Mathematics*, vol. 6, p. 104, Jun 2018.